

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5111

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks:50

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

*I Kala Gowri Shankar(192325109) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Kala Gowri Shankar

Annexure A

TECHNICAL PRESENTATION

1. From Caesar to Vigenère: Evolution of Substitution Ciphers”
2. “Substitution vs Transposition: Which Provides Better Security?”
3. “Security Attacks, Threats, Risks and Their Impact on Security Goals”
4. “Steganography vs Cryptography: Hidden Security in the Digital World”
5. “Role of Number Theory in Cryptography: From Euclid to Euler”

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, wellstructured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation



Public-Key Cryptograph

Public-key cryptography, also known as asymmetric encryption, is a secure method of protecting digital information using two different keys: a public key and a private key. Unlike symmetric encryption, which uses the same key for both encryption and decryption, asymmetric encryption uses a public key that can be shared openly and a private key that is kept secret by its owner. The public key is used to encrypt a message, while only the corresponding private key can decrypt it. This approach provides strong security for communication over the internet because the private key never needs to be shared. The process begins with the generation of a key pair consisting of a public key and a matching private key. The public key is then distributed openly so that anyone can use it to send encrypted messages to the owner. When a sender wants to transmit confidential information, the message is encrypted using the receiver's public key. Once encrypted, only the receiver's private key can decrypt the message and reveal its original content, ensuring that unauthorized users cannot access the information. Public-key cryptography also enables digital signatures, which verify the authenticity and integrity of digital messages. A sender creates a digital signature using their private key, and anyone with the sender's public key can verify that the signature is genuine. If the message is altered after signing, the signature becomes invalid, making it easy to detect tampering. This feature is widely used to establish trust in online communications.